

JEE MAIN REVISION MASTER: ALTERNATING CURRENT

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 32 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Series LCR Resonance, Power Factor & Transformers	17 PYQs	Resonance & Transformers
Very High	Peak, RMS Values & AC Through R, L, C	15 PYQs	AC Fundamentals

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Series LCR Resonance, Power Factor & Transformers	$f_0 = \frac{1}{2\pi\sqrt{LC}}$, $Z = \sqrt{R^2 + (X_L - X_C)^2}$, $\cos\phi = \frac{R}{Z}$, $\frac{V_s}{V_p} = \frac{N_s}{N_p}$
Peak, RMS Values & AC Through R, L, C	$I_{\text{rms}} = \frac{I_0}{\sqrt{2}}$, $X_L = 2\pi f L$, $X_C = \frac{1}{2\pi f C}$, $P_{\text{avg, pure L/C}} = 0$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Series LCR Resonance, Power Factor & Transformers (Very High Yield — 17 PYQs)

Core Principle: Series LCR impedance $Z = \sqrt{R^2 + (X_L - X_C)^2}$. At resonance ($X_L = X_C$): $Z = R$ (minimum), current is maximum, resonant frequency $\omega_0 = \frac{1}{\sqrt{LC}}$, and power factor $\cos\phi = 1$. Power dissipated: $P = V_{\text{rms}} I_{\text{rms}} \cos\phi$. Ideal transformer: $(V_s / V_p) = (N_s / N_p) = (I_p / I_s)$.

■ **Caution / JEE Pitfall:** Voltages across L and C in a resonant circuit can individually exceed the applied source voltage because V_L and V_C are 180° out of phase and cancel out!

■ **Speed Trick:** Step-up transformer increases voltage but decreases current ($I_s < I_p$) in accordance with energy conservation ($V_p I_p = V_s I_s$).

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "A small town with a demand of 900kW of electric power at 220V is situated 20km away from an electric power genera..." [Ans: N/A]
- [2025 Shift] Q#2 (MCQ): "A series LCR circuit containing an AC source of 100 V has an inductor and a capacitor of reactances 24Ω and 16Ω respectively. ..." [Ans: N/A]
- [2024 Shift] Q#1 (MCQ): "The figure shows variation of R, X_L and X_C with frequency f in a series L-C-R circuit. Then, for what frequency point is the ci..." [Ans: N/A]

3.2 Peak, RMS Values & AC Through R, L, C (Very High Yield — 15 PYQs)

Core Principle: AC relations: $I_{\text{rms}} = \frac{I_0}{\sqrt{2}} \approx 0.707 I_0$, $V_{\text{rms}} = \frac{V_0}{\sqrt{2}}$. Inductive reactance $X_L = \omega L = 2\pi f L$ (current lags voltage by 90°). Capacitive reactance $X_C = (1 / \omega C) = (1 / 2\pi f C)$ (current leads voltage by 90°). For pure L or C, average power consumed over a cycle is strictly ZERO.

■ **Caution / JEE Pitfall:** Household AC supply specification 220 V is the RMS voltage; peak voltage is $V_0 = 220\sqrt{2} \approx 311 \text{ V}$!

■ **Speed Trick:** In a DC circuit ($f = 0$): $X_L = 0$ (inductor acts as short circuit / straight wire) and $X_C = \infty$ (capacitor acts as open circuit / blocks DC).

Representative Source PYQs:

- **[2026 Shift] Q#2 (MCQ):** "A light bulb rated 100 W is connected to an AC source of 220 V, 50 Hz. The rms current through the bulb is..." **[Ans: N/A]**
- **[2026 Shift] Q#3 (MCQ):** "In a circuit containing a pure resistor connected to an AC source,..." **[Ans: N/A]**
- **[2025 Shift] Q#1 (MCQ):** "A sinusoidal voltage produced by an AC generator at any instant t is given by an equation $V = 311 \sin 314 t$. The rms value of voltage and freq..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

A small town with a demand of 900kW of electric power at 220V is situated 20km away from an electric power generating station. Each wire of the two-wire transmission line has a resistance per unit length of $5 \times 10^{-4} \Omega \text{m}^{-1}$. The town gets power from the line through a 45000V to 220V step-down transformer at a substation in the town. The line power loss in the form of heat is:

- (A) 4 kW
- (B) 8 kW
- (C) 40 kW
- (D) 80 kW

Correct Answer: None

Solution Analysis: The town receives power through a step-down transformer whose primary voltage is 45000V. The power demand of the town is $P = 900 \text{ kW} = 900000 \text{ W}$. So the current in the transmission line is $I = (P / V) = (900000 / 45000) = 20 \text{ A}$. Now, each wire is 20km long. $20 \text{ km} = 20000 \text{ m}$. Resistance per unit length of each wire is $5 \times 10^{-4} \Omega \text{m}^{-1}$. So resistance of one wire is $R = 5 \times 10^{-4} \times 20000 = 10 \Omega$. Since it is a two-wire transmission line, total resistance is $R_{\text{total}} = 10 + 10 = 20 \Omega$. Power loss in the line

Benchmark Problem 2: KCET 2026 (Q#2)

A light bulb rated 100 W is connected to an AC source of 220 V, 50 Hz. The rms current through the bulb is

- (A) 0.454 A
- (B) 0.545 A
- (C) 2.20 A
- (D) 0.22 A

Correct Answer: None

Solution Analysis: For a light bulb, the given power rating is $P=100\text{W}$ and the AC voltage given is the rms voltage, $V_{\text{(rms)}}=220\text{V}$. For a bulb, which behaves like a resistor, the power is $P=V_{\text{(rms)}}I_{\text{(rms)}}$. So, $I_{\text{(rms)}}=\frac{P}{V_{\text{(rms)}}}$. Substitute the values: $I_{\text{(rms)}}=(100 / 220)$ $I_{\text{(rms)}}=0.454\text{A}$. So, the rms current through the bulb is 0.454A. Correct option: Option A.